Origin of the Generator $H^{(l)}$ and the Matrix Exponential in Rotational Diffusion

1 Introduction

In the theory of rotational diffusion one frequently encounters the relation

$$G^{(l)}(t) = e^{-H^{(l)}t},$$

where $G^{(l)}(t)$ is the propagator in the rank-l Wigner basis and $H^{(l)}$ is a finite-dimensional matrix called the rotational diffusion generator.

The purpose of this note is to explain carefully:

1. where the matrix $H^{(l)}$ comes from,
2. why the propagator becomes a matrix exponential,
3. how the infinite-dimensional diffusion equation reduces to a finite matrix problem for fixed angular momentum rank l.

2 Rotational diffusion equation

The orientation of a rigid body is described by Euler angles

$$\Omega = (\alpha, \beta, \gamma).$$

The conditional probability density

$$P(\Omega, t \mid \Omega_0, 0)$$

gives the probability density of finding the molecule at orientation Ω at time t, given that it started at orientation Ω_0 at time 0.

Rotational Brownian motion is governed by the rotational Smoluchowski equation

$$\frac{\partial}{\partial t} P(\Omega, t \mid \Omega_0, 0) = -\hat{\Gamma}\, P(\Omega, t \mid \Omega_0, 0),$$

with initial condition

$$P(\Omega, 0 \mid \Omega_0, 0) = \delta(\Omega - \Omega_0).$$

For anisotropic rotational diffusion the diffusion operator is

$$\hat{\Gamma} = D_x \hat{J}_x^2 + D_y \hat{J}_y^2 + D_z \hat{J}_z^2,$$

where $\hat{J}_x, \hat{J}_y, \hat{J}_z$ are angular momentum operators acting on functions defined on the rotation group $SO(3)$, and D_x, D_y, D_z are the principal-axis rotational diffusion constants.

This equation is the rotational analogue of the translational diffusion equation

$$\frac{\partial P}{\partial t} = D\nabla^2 P.$$

3 Wigner functions as a basis

The natural basis functions on $SO(3)$ are the Wigner rotation matrix elements

$$D^l_{mn}(\Omega).$$

These satisfy the orthogonality relation

$$\int d\Omega\, D^{l*}_{mn}(\Omega)\, D^{l'}_{m'n'}(\Omega) = \frac{8\pi^2}{2l+1}\delta_{ll'}\delta_{mm'}\delta_{nn'}.$$

The key fact is that the diffusion operator preserves angular momentum rank l. In other words, when $\hat{\Gamma}$ acts on a Wigner function of rank l, the result is always a linear combination of Wigner functions of the same rank:

$$\hat{\Gamma} D^l_{mn} = \sum_{n'} H^{(l)}_{nn'} D^l_{mn'}.$$

This equation defines the matrix $H^{(l)}$.

The operator $\hat{\Gamma}$ acts on an infinite-dimensional function space, but once we restrict to fixed rank l, the problem reduces to a finite-dimensional vector space of dimension $2l+1$.

For fixed l, the functions

$$\{D^l_{m,-l}, \ldots, D^l_{ml}\}$$

form a basis. The diffusion operator therefore becomes an ordinary matrix.

4 How the matrix elements are obtained

The matrix elements of $H^{(l)}$ are simply the representation of the operator $\hat{\Gamma}$ in the Wigner basis:

$$H^{(l)}_{nn'} = \frac{2l+1}{8\pi^2} \int d\Omega\, D^{l*}_{mn}(\Omega)\, \hat{\Gamma} D^l_{mn'}(\Omega).$$

The matrix is independent of m because the diffusion operator acts only on the body-fixed index.

For anisotropic diffusion,

$$\hat{\Gamma} = D_x \hat{J}_x^2 + D_y \hat{J}_y^2 + D_z \hat{J}_z^2,$$

so

$$H^{(l)} = D_x J_x^{(l)2} + D_y J_y^{(l)2} + D_z J_z^{(l)2},$$

where $J_i^{(l)}$ are the ordinary angular momentum matrices for angular momentum l.

Thus $H^{(l)}$ is nothing more than the matrix representation of the rotational diffusion operator in the rank-l irreducible representation of $SO(3)$.

5 Example: rank $l = 2$

For $l = 2$, the space has dimension 5, corresponding to

$$m = -2, -1, 0, 1, 2.$$

In the ordered basis

$$(2, 1, 0, -1, -2),$$

the generator becomes the 5×5 matrix

$$H = \begin{pmatrix} A & 0 & U & 0 & 0 \\ 0 & B & 0 & V & 0 \\ U & 0 & C & 0 & U \\ 0 & V & 0 & B & 0 \\ 0 & 0 & U & 0 & A \end{pmatrix},$$

where

$$A = D_x + D_y + 4D_z,$$

$$B = \frac{5}{2}(D_x + D_y) + D_z,$$

$$C = 3(D_x + D_y),$$

$$U = \frac{\sqrt{6}}{2}(D_x - D_y),$$

$$V = \frac{3}{2}(D_x - D_y).$$

6 Why the propagator is a matrix exponential

Suppose we expand the propagator in the Wigner basis:

$$P(\Omega, t \mid \Omega_0, 0) = \sum_{nn'} D^l_{mn}(\Omega)\, G^{(l)}_{nn'}(t)\, D^{l*}_{mn'}(\Omega_0).$$

Substituting this expansion into the diffusion equation gives

$$\frac{d}{dt} G^{(l)}(t) = -H^{(l)} G^{(l)}(t).$$

This is now an ordinary matrix differential equation.
The initial condition comes from

$$P(\Omega, 0 \mid \Omega_0, 0) = \delta(\Omega - \Omega_0),$$

which implies

$$G^{(l)}(0) = I.$$

Therefore the solution is

$$\boxed{G^{(l)}(t) = e^{-H^{(l)}t}.}$$

This is exactly analogous to the scalar ordinary differential equation

$$\frac{dx}{dt} = -ax,$$

whose solution is

$$x(t) = e^{-at} x(0).$$

The matrix exponential is defined by the convergent power series

$$e^{-H^{(l)}t} = \sum_{n=0}^{\infty} \frac{(-t)^n}{n!} (H^{(l)})^n.$$

7 Physical interpretation

The eigenvalues of $H^{(l)}$ are rotational relaxation rates.

If

$$H^{(l)} v_i = \lambda_i v_i,$$

then each eigenmode decays exponentially:

$$e^{-H^{(l)} t} v_i = e^{-\lambda_i t} v_i.$$

Thus anisotropic rotational diffusion is ultimately a superposition of exponential relaxation modes.

In the isotropic case,

$$D_x = D_y = D_z = D_R,$$

the generator becomes proportional to the identity:

$$H^{(l)} = l(l+1) D_R \, I,$$

and therefore

$$G^{(l)}(t) = e^{-l(l+1) D_R t} I.$$

For $l = 2$,

$$G^{(2)}(t) = e^{-6 D_R t} I_5.$$